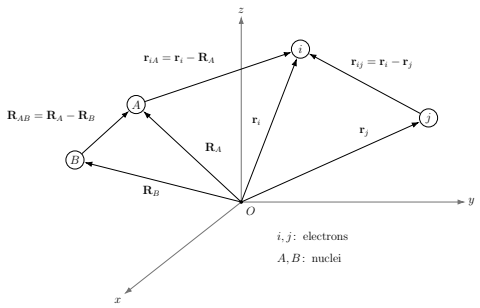
1. Molecular coordinates

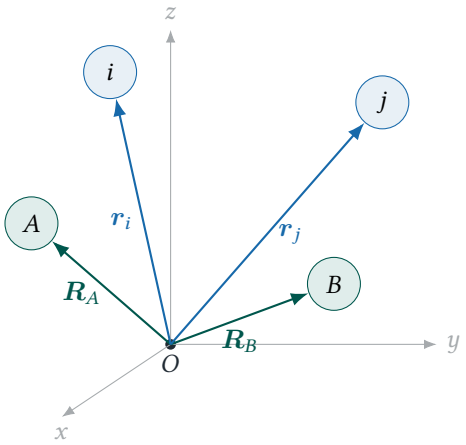
Proposed replacement for current Fig. 1

Current illustration

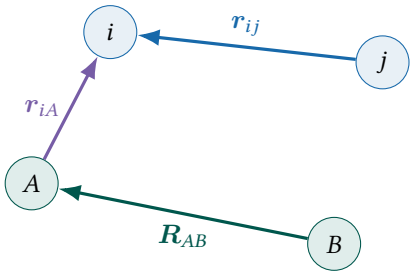


Proposed design: separate positions from displacements

Position vectors



Relative vectors



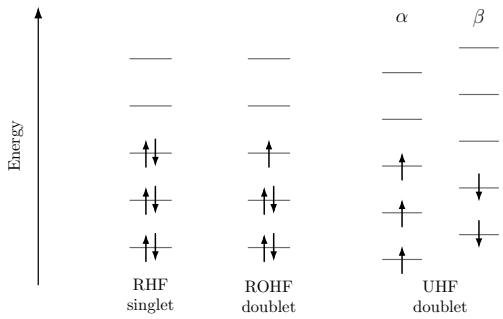
$$\boldsymbol{r}_{iA} = \boldsymbol{r}_i - \boldsymbol{R}_A \qquad \boldsymbol{r}_{ij} = \boldsymbol{r}_i - \boldsymbol{r}_j \qquad \boldsymbol{R}_{AB} = \boldsymbol{R}_A - \boldsymbol{R}_B$$

Nuclei: green; electrons: blue. The two panels use the same geometry. New vector drawing following the coordinate definitions in Szabo and Ostlund, *Modern Quantum Chemistry*. The source credit will remain.

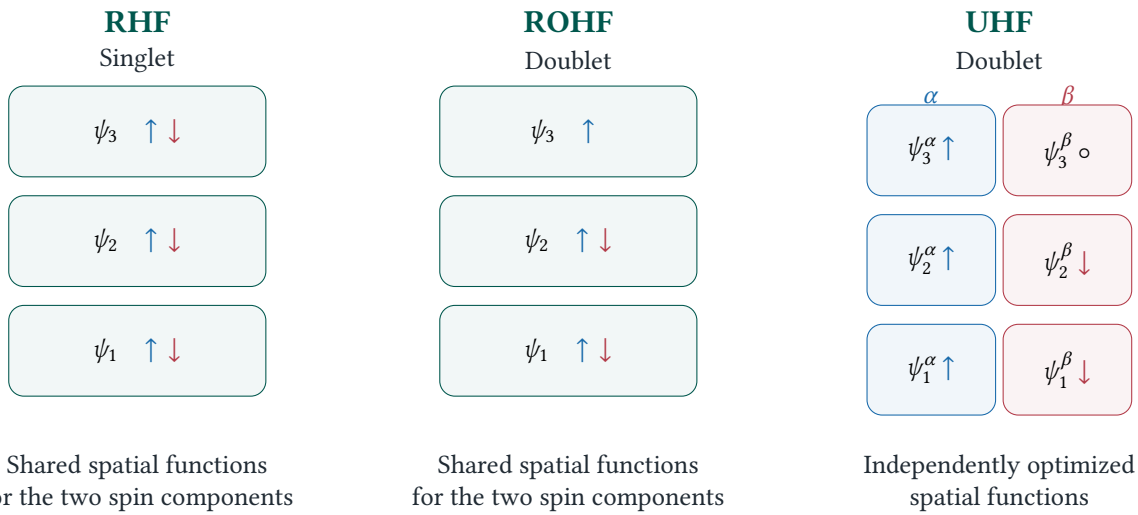
2. Restricted and unrestricted orbitals

Proposed replacement for current Fig. 2

Current illustration



Proposed design: show which spatial functions are shared

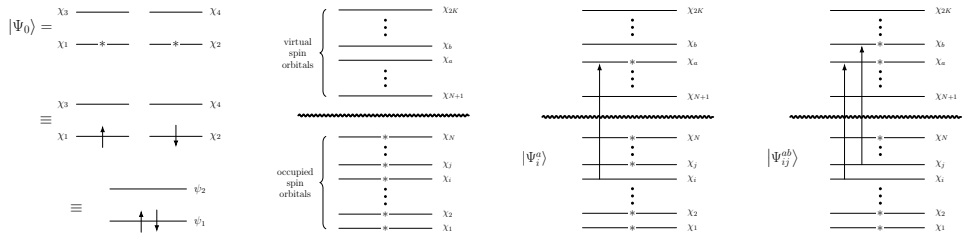


Illustrative occupations retain the singlet/doublet examples and the ROHF case in the source. This design emphasizes orbital sharing rather than orbital-energy order; it is not an energy-level plot. New schematic following Jensen, *Introduction to Computational Chemistry*.

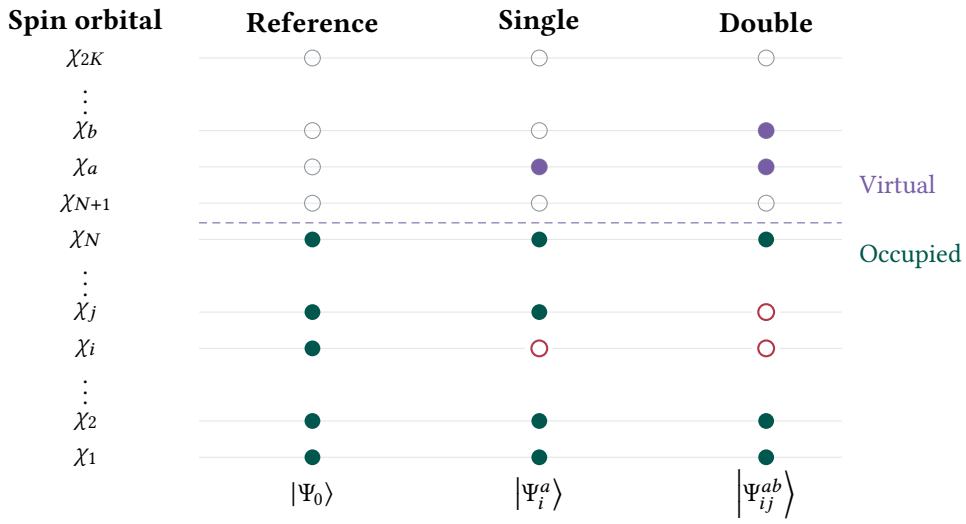
3. Determinant occupations

Proposed coordinated designs for current Figs. 3, 4, and 5

Current occupation diagrams



Proposed design: occupation maps in a common orbital basis



Filled green: an unchanged occupied spin orbital; violet: a promoted electron; red outline: a hole. Empty cells are unoccupied. Unshown occupations remain unchanged. Rows show orbital labels, not numerical energies.

Closed-shell H_2: the same graphic vocabulary

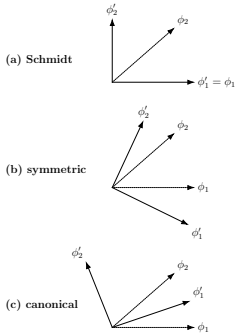
$$\underbrace{\chi_1 = \alpha\psi_1, \quad \chi_2 = \beta\psi_1}_{\text{occupied}} \qquad \underbrace{\chi_3 = \alpha\psi_2, \quad \chi_4 = \beta\psi_2}_{\text{virtual}} \qquad \Longleftrightarrow \qquad \psi_1^2$$

New occupation-map layout following the determinant constructions of Szabo and Ostlund, *Modern Quantum Chemistry*. The source credit is retained.

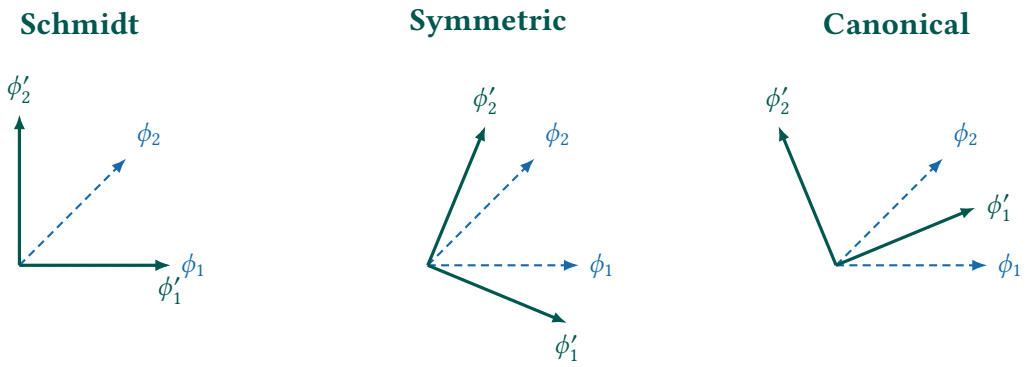
4. Orthogonalization

Proposed replacement for current Fig. 6

Current illustration



Proposed design: parallel method panels and transformation rules



Sequential

Retain ϕ_1; subtract its component from ϕ_2 and normalize.

Equal treatment

$\boldsymbol{X} = \boldsymbol{S}^{-1/2}$

Overlap eigenvectors

$\boldsymbol{X} = \boldsymbol{U} \cdot \boldsymbol{s}^{-1/2}$

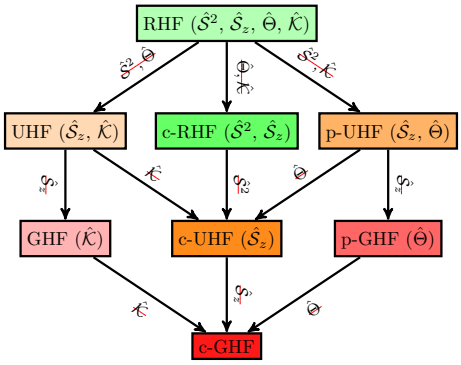
$$\boldsymbol{X}^{\dagger} \cdot \boldsymbol{S} \cdot \boldsymbol{X} = \mathbf{1}$$

Dashed blue: original functions; solid green: orthonormal functions. An illustrative pair of real normalized functions is used. Each green pair is perpendicular; the methods choose different orthonormal coordinates in the same space. Independently drawn panels following Szabo and Ostlund.

5. HF symmetry classes

Proposed replacement for current Fig. 10

Current illustration



Proposed design: constraint matrix plus explicit release map

Space	Real coefficients	Time-reversal paired	Complex coefficients
RHF	RHF: $\hat{S}^2, \hat{S}_z, \hat{\mathcal{K}}, \hat{\Theta}$	Not a separate class here	c-RHF: $\hat{S}^2, \hat{S}_z$
UHF	UHF: $\hat{S}_z, \hat{\mathcal{K}}$	p-UHF: $\hat{S}_z, \hat{\Theta}$	c-UHF: $\hat{S}_z$
GHF	GHF: $\hat{\mathcal{K}}$	p-GHF: $\hat{\Theta}$	c-GHF: none of these

Cells name the retained symmetries. Real and paired classes are alternative constraints, not consecutive steps in an ordered chain.

All constraint-release connections retained from the original graph

RHF → c-RHF: release $\hat{\Theta}, \hat{\mathcal{K}}$

RHF → p-UHF: release $\hat{S}^2, \hat{\mathcal{K}}$

UHF → GHF: release $\hat{S}_z$

p-UHF → p-GHF: release $\hat{S}_z$

c-UHF → c-GHF: release $\hat{S}_z$

p-GHF → c-GHF: release $\hat{\Theta}$

RHF → UHF: release $\hat{S}^2, \hat{\Theta}$

c-RHF → c-UHF: release $\hat{S}^2$

UHF → c-UHF: release $\hat{\mathcal{K}}$

p-UHF → c-UHF: release $\hat{\Theta}$

GHF → c-GHF: release $\hat{\mathcal{K}}$

New comparison layout based on Fig. 1 of Burton, Thom, and Loos, *Parity-Time Symmetry in Hartree-Fock Theory* (2019), doi:10.1021/acs.jctc.9b00289. The chapter's existing Table 1 remains a separate, credited reproduction in part from Henderson, Jimenez-Hoyos, and Scuseria (2018).

6. Carbon contraction data

Proposed table replacing current Fig. 12

Current image excerpt

```
C     0
S    10   1.00
      8.236000D+03          5.310000D-04
      1.235000D+03          4.108000D-03
      2.808000D+02          2.108700D-02
      7.927000D+01          8.185300D-02
      2.559000D+01          2.348170D-01
      8.997000D+00          4.344010D-01
      3.319000D+00          3.461290D-01
      9.059000D-01          3.937800D-02
      3.643000D-01         -8.983000D-03
      1.285000D-01          2.385000D-03
```

Proposed native LaTeX table

Primitive i	Exponent α_i	Coefficient D_i
1	8.236000×10^{3}	5.310000×10^{-4}
2	1.235000×10^{3}	4.108000×10^{-3}
3	2.808000×10^{2}	2.108700×10^{-2}
4	7.927000×10^{1}	8.185300×10^{-2}
5	2.559000×10^{1}	2.348170×10^{-1}
6	8.997000×10^{0}	4.344010×10^{-1}
7	3.319000×10^{0}	3.461290×10^{-1}
8	9.059000×10^{-1}	3.937800×10^{-2}
9	3.643000×10^{-1}	-8.983000×10^{-3}
10	1.285000×10^{-1}	2.385000×10^{-3}

Proposed caption. Primitive exponents and coefficients of one ten-primitive s contraction for carbon in cc-pVTZ. Shell scale: 1.00. Values transcribed from the lecture excerpt and checked against the Basis Set Exchange; basis family: Dunning (1989), doi:10.1063/1.456153.

Only the complete ten-row contraction visible in the image is included. Every exponent and coefficient, including the negative ninth coefficient, is retained exactly. The raw D-format notation is replaced by standard scientific notation; the values have not been rounded or renormalized.

Approval pending. The chapter still contains the original Fig. 12. This table and all proposed redraws are separate review material. No replacement will be incorporated until you approve it.